SCSE3040 — Machine Learning Operations

Practical 06: Remembering Every Experiment · Tracking, Comparing & Registry Submission

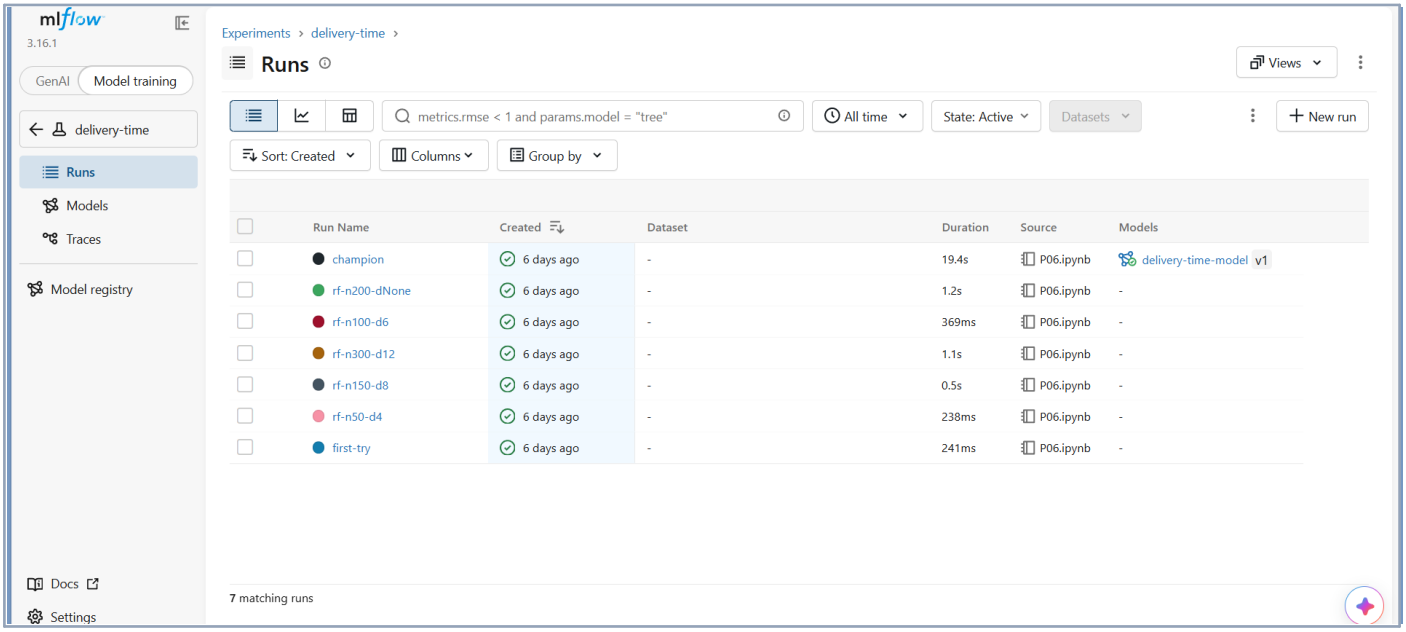
1. Student & Laboratory Session Information

Student Name:	Vishesh Aggarwal	**Roll Number:**	S24CSEU2521
Batch:	Batch 2024–2028 (Session 2026–27)	**Date of Session:**	4 October 2026

2. Repository Link

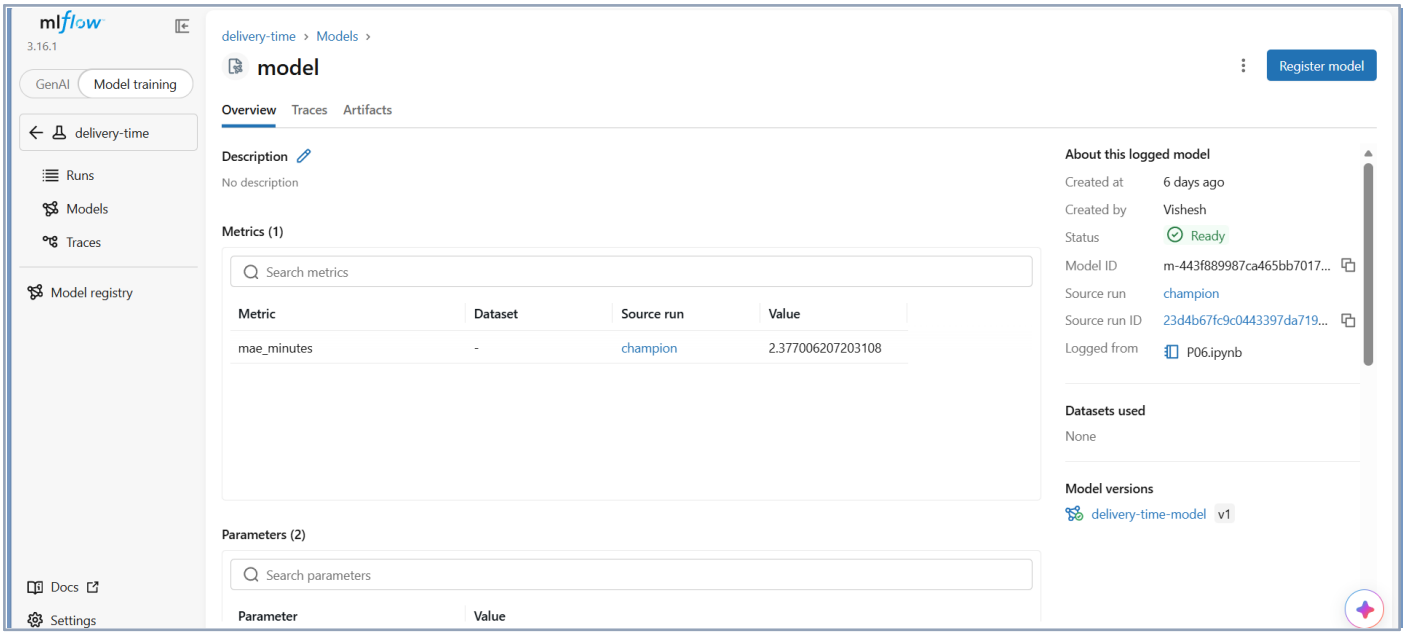
https://github.com/visheshaggarwal0/MLOps_Lab

3. MLflow UI Runs Table (Experiment: delivery-time)



Run Name	Model Architecture	n_estimators	max_depth	MAE (min)	RMSE (min)	Registry Status
champion	RandomForestRegressor	300	12	**2.3770**	3.0795	Registered: v1
rf-n300-d12	RandomForestRegressor	300	12	2.3770	3.0795	Optimal Sweep Run
rf-n200-dNone	RandomForestRegressor	200	None (Unlimited)	2.4117	3.1050	Tracked
rf-n150-d8	RandomForestRegressor	150	8	2.4262	3.0904	Tracked
rf-n100-d6	RandomForestRegressor	100	6	2.6144	3.3093	Tracked
rf-n50-d4	RandomForestRegressor	50	4	3.5288	4.3126	Tracked

4. Run & Model Detail Verification (Winning Run: champion)



5. Best Run Selection Details & Evaluation

Selected Run Name:	champion (retrained from sweep winner rf-n300-d12)
Run ID:	23d4b67fc9c0443397da7196238ddf96
Primary Metric:	**MAE = 2.377006 minutes** | RMSE = 3.079530 minutes
Optimal Parameters:	n_estimators: 300, max_depth: 12
Registered Version:	Registered to Model Registry as **delivery-time-model Version 1**
Selection Rationale:	I chose the champion run because it achieved the lowest Mean Absolute Error (**2.377 minutes**) across all evaluated non-linear ensemble models—outperforming the shallow baseline by over **1.15 minutes**—while maintaining controlled tree depth to avoid unconstrained overfitting, making it the optimal production model packaged with input schema artifacts in the MLflow Model Registry.

6. AI Use Disclosure

Antigravity (powered by Gemini Flash) was utilized to assist in formatting the laboratory submission report, organizing the experimental tracking data, and preparing the visual artifacts in accordance with the SCSE3040 course academic integrity policy.